Pool-ball localisation and recognition

Manon Yiqin Galian Ng (23375206)

1. Finding the balls

The system finds possible balls, recognises them and converts their centres to table coordinates. AllBalls supplies training examples; Table1–Table6 contain 90 annotated balls. One parameter set processes every image. Later rules followed inspection of these six images, so the scores are not an independent test on new scenes.

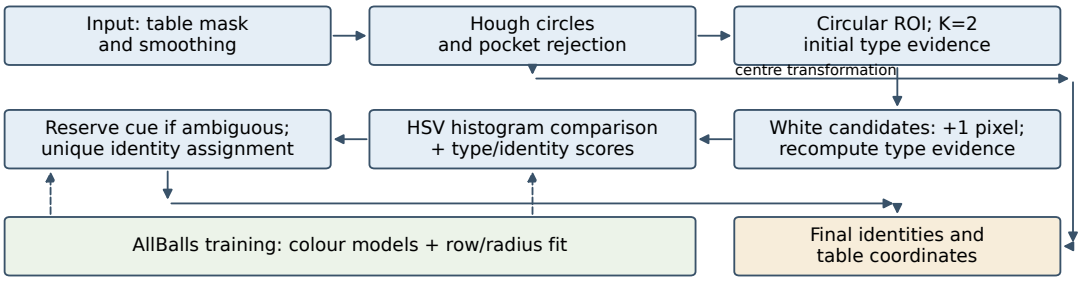


Figure 1: The processing stages. Dashed arrows supply AllBalls models. Position conversion is separate from recognition.

Masking and circle detection

The camera does not move, so the supplied table corners define a fixed playing area: $(337, 35)$, $(1570, 21)$, $(1888, 451)$ and $(38, 472)$, written as $(\text{column}, \text{row})$. Pixels outside this quadrilateral are masked out. This prevents the surrounding room from producing ball detections. A 5×5 Gaussian filter, with standard deviation 1.5 pixels, reduces noise in the greyscale image before circular Hough detection.

Hough settings were compared on AllBalls in a 480-combination search. The selected accumulator resolution ratio is 1.5, minimum centre spacing 18 pixels, upper edge threshold 75 and circle-voting threshold 35; allowed radii are 15–30 pixels. The circle estimates are candidates: pockets and parts of neighbouring balls can also produce circular edges.

Rejecting pockets

Candidate centres are converted to table coordinates and compared with six known pocket positions. A candidate within 25 table units of a pocket centre is rejected. This uses the table layout to remove predictable false detections, but could also remove a genuine ball close to a pocket.



Figure 2: AllBalls detection stages. Cyan guides mark centre rows used to estimate radius. Recognition uses original colours.

2. Recognising ball type and colour

Each detected circle defines a square region of interest (ROI). A circular mask restricts recognition to pixels inside the estimated ball. The system asks two separate questions: *how much of the ball appears white?* and *which body colour does it resemble?* This separates type (cue, solid or striped) from colour, since solid and striped partners share a colour.

K-means estimates the amount of white

K-means divides the ROI's BGR pixels into two colour groups. Pixels in a group need not touch each other. Each group is represented by its average colour, which is converted to HSV. Saturation S describes colour strength; value V describes brightness. Both use the range 0–255 here.

A group is light if $V \geq 170$ and $S \leq 165$. Otherwise it is dark if $V \leq 90$, and coloured otherwise. The sizes of these groups give light and non-light percentages. At least 85% light initially suggests cue; at least 70% non-light suggests solid; a remaining mixture suggests stripe. Dark pixels count as non-light so dark ball bodies are not discarded. Clustering uses three initialisations, a fixed seed, and at most 50 iterations or a centre change below 0.2. It reduces sensitivity to individual pixels, but may merge a pale body colour with white.

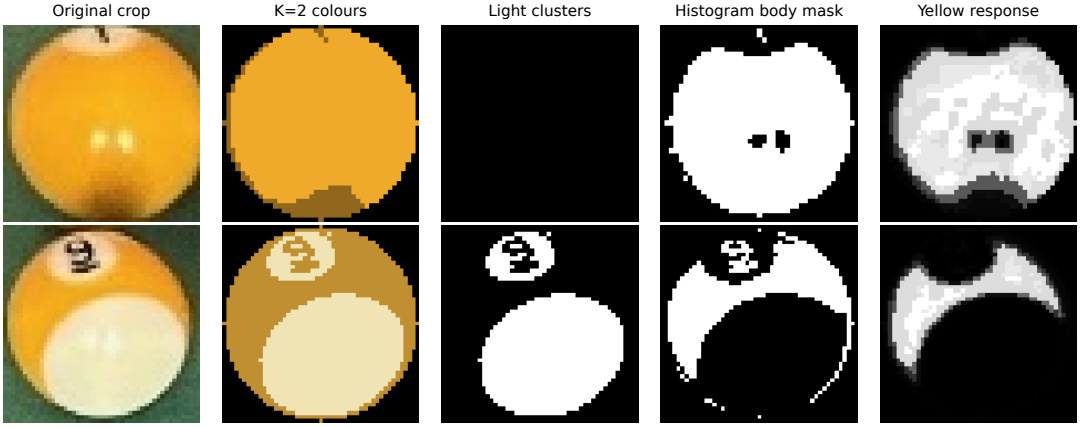


Figure 3: Solid 1 (top) and striped 9 (bottom). The light-cluster mask estimates white coverage; the body mask supplies colour samples.

HSV histograms identify the body colour

AllBalls supplies eight colour models: yellow, blue, red, purple, orange, green, maroon and black. Solid/striped partners contribute equally to their shared model; cue is excluded. Before measuring body colour, the circular mask is eroded with a 3×3 ellipse and light pixels are removed. This reduces white-marking and background influence. Fewer than 12 remaining pixels gives no usable body-colour evidence.

Each histogram counts pixels in $18 \times 8 \times 8$ hue/saturation/value bins, then divides by its total. The bin ranges are $[0, 180)$, $[0, 256)$ and $[0, 256)$. Neighbouring bins are smoothed with weights $(1, 2, 1)/4$; hue wraps around. For nearly grey pixels (BGR channel range below 15), hue is set to zero. A crop is compared with each trained histogram using one minus Bhattacharyya distance: a higher score means a closer colour distribution [1]. Backprojection shows where a model responds in the crop; its brightness is scaled for display and is not the score used to choose an identity.

3. Combining evidence and producing positions

Avoiding conflicting identities

The initial type is useful but can be wrong near a threshold. For example, a mostly white stripe may be labelled cue. We therefore keep scores for possible identities instead of immediately accepting one label. A numbered-ball score multiplies colour similarity by agreement with the expected white coverage: cue 85–100%, solid 0–30%, stripe 30–85%. Agreement is highest inside its range and falls smoothly outside it, using a Gaussian width of 0.10 in light fraction. Cue's score uses light fraction in place of histogram similarity; black supports ball 8 only. These scores are not calibrated probabilities.

The Hungarian method [2] maximises total score while using each identity at most once. If two candidates prefer cue, one may receive its next-best striped identity. Balls need not all be present. Unknown has effectively zero score and is available when evidence is missing or uniqueness leaves no supported choice. Receiving an identity requires usable body colour or an initial cue label. Unknown retains the detection, including a false circle.

Padding white candidates and selecting the cue

Hough sometimes follows a stripe's inner white region rather than its full boundary. Every ROI initially labelled cue is therefore expanded by one pixel, and its type and histogram evidence are recomputed. This may expose a coloured rim, but also admits cloth. Other ROIs are unchanged.

The full cue should generally be larger than an internal white patch, after accounting for depth. A median-slope line [3] fits the 16 AllBalls centre rows and radii (Appendix B):

$$\hat{r}(y) = \max(1, 20.274 + 0.009677y).$$

With at least two original cue-like candidates, cue is reserved for the largest original-radius-to-expected-radius ratio, $r_i/\hat{r}(y_i)$. Padding does not increase this ratio. This is a forced rule, not proof that the chosen ball is cue.

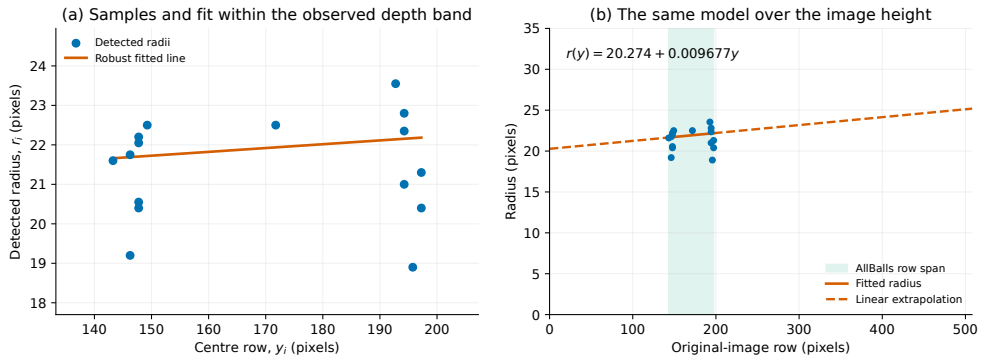


Figure 4: The radius model uses a narrow training band. Dashed sections estimate sizes beyond the observed rows. Source measurements appear in Appendix B.

Converting centres to table coordinates

A homography maps the supplied corners onto a 1434×683 image, matching the table dimensions in millimetres. The same perspective correction converts detected centres. Pixel endpoints are 0–1433 and 0–682. Recognition leaves positions unchanged. Appendix A shows the final views.

4. Measuring performance

Predicted centres are matched one-to-one with annotated centres within 15 table units, without using ball identities. The matching finds as many valid pairs as possible, considering nearby alternatives first. A matched detection is a true positive (TP); an unmatched detection is a false positive (FP); an unmatched annotation is a false negative (FN). The matching does not guarantee the lowest possible total position error in a crowded group.

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F1 = \frac{2PR}{P + R}.$$

Precision measures how many detections are real balls; recall measures how many annotated balls were found. Mean position error averages distances over matched pairs. Identity accuracy counts correct identities among those matches. Unknown counts as an error on a genuine ball. End-to-end identification also includes missed balls, using all annotations as its denominator.

Table 1: Results for each image. Error is in table units (approximately mm); identity accuracy considers matched detections only.

Image	GT	TP	FP	FN	Error	Correct ID	ID acc.
Table1	16	14	1	2	7.68	11/14	78.57%
Table2	16	16	1	0	3.58	16/16	100.00%
Table3	16	15	0	1	3.12	15/15	100.00%
Table4	15	14	0	1	3.89	14/14	100.00%
Table5	14	14	0	0	5.83	13/14	92.86%
Table6	13	13	0	0	5.76	11/13	84.62%
ALL	90	86	2	4	4.91	80/86	93.02%

Detection precision is 86/88 = 97.73%, recall 86/90 = 95.56% and F1 96.63%. Mean error is 4.91 units, averaged over all 86 matches. Type and identity totals both equal 80/86 (93.02%). End-to-end identification is 80/90 = 88.89%. The two Unknowns are Table1's real stripe 14 and Table2's unmatched extra circle. AllBalls obtains 16/16 identities, but it supplied the training examples, so this is not an independent test result.

Table 2: Adding recognition stages while keeping the same detections.

Configuration	Correct identities
K=2 with fixed colour rules	68/86
K=2 with HSV histograms	72/86
Add unique assignment	75/86
Add selective +1 pixel padding	79/86
Reserve largest depth-adjusted cue	80/86

These gains change recognition, not detection counts. The same settings were used throughout each run, and annotations were loaded after predictions were made. However, choosing rules after inspecting Table1–Table6 still influences the method. New images would be needed to estimate performance independently.

5. Explaining the results

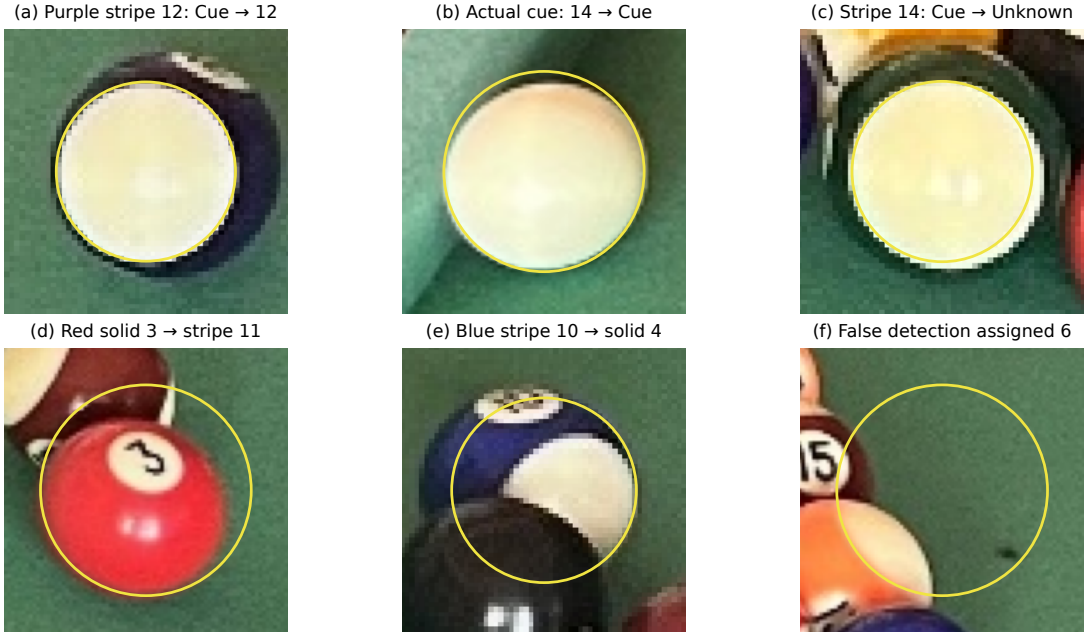


Figure 5: Examples of corrections and remaining errors. Arrows show earlier-to-selected identities; circles show original detections.

Padding corrects cue/stripe conflicts in Tables3 and 6. Extra colour lets the striped candidate take another identity, releasing cue for the real cue ball. Two corrected pairs give four correct identities. The crop alone may still prefer cue; the joint assignment resolves the conflict.

Table1's real cue includes misleading green cloth. Stripe 14 has only eight usable coloured pixels after padding, below the required 12. Reserving cue restores the real cue but leaves stripe 14 Unknown. Table1 also swaps solid 5 and stripe 9.

Tables5 and 6 confuse blue stripes with solid 4; Table6 calls solid 3 stripe 11. Orientation and highlights change white coverage; inaccurate circles include cloth or neighbours. Uniqueness can also create wrong swaps or give a false circle a plausible identity (Figure 5(f)).

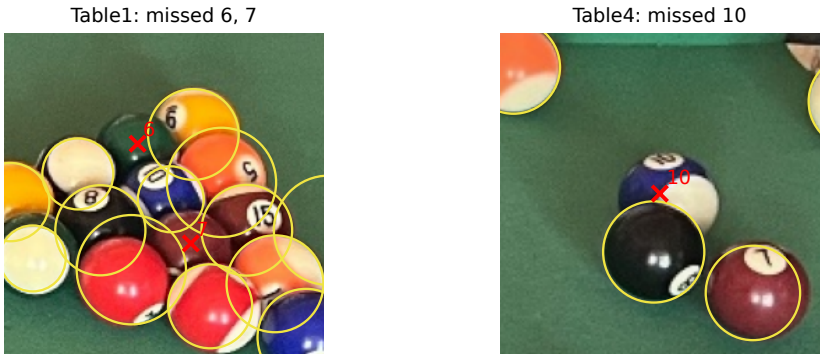


Figure 6: Yellow circles are detections; red crosses are supplied centres projected into the source image.

Four balls are missed: 6 and 7 (Table1), and 10 (Tables3/4). Occlusion and merged boundaries weaken circular evidence; minimum spacing may suppress close centres.

6. Alternatives and limitations

Why additional segmentation was not retained

Watershed separates regions by growing them from starting markers [4]. It was tested to separate overlapping balls. Starting from Hough markers did not recover any misses: a missing ball often also lacked a useful marker. Finding clusters first and applying more permissive local Hough searches recovered three annotations, but lost three previous matches and increased false positives from 2 to 16. F1 fell from 96.63% to 89.58%.

Connected-component analysis (CCA) measured connected white/coloured areas and how much of the crop boundary they reached. The aim was to distinguish a stripe from a small white number patch. With the same HSV colour models, the tested rules reduced correct identities from 72 to 69. White-region shape changes with orientation, and highlights or shadows can split one region into several.

Chamfer matching compares a proposed circle boundary with nearby image edges [5]. Searching centre shifts up to two pixels and radius growth up to eight fixed the Table1 cue/14 swap when an outer supported circle was preferred. However, it scored 77/86 overall, below padding's 79/86. A one-pixel shift exposed 15 body pixels instead of eight: a useful correction, but sensitive to crop placement and the 12-pixel cutoff. Direct HSV pixel counting also scored 79/86 in the final combination, compared with K-means' 80/86.

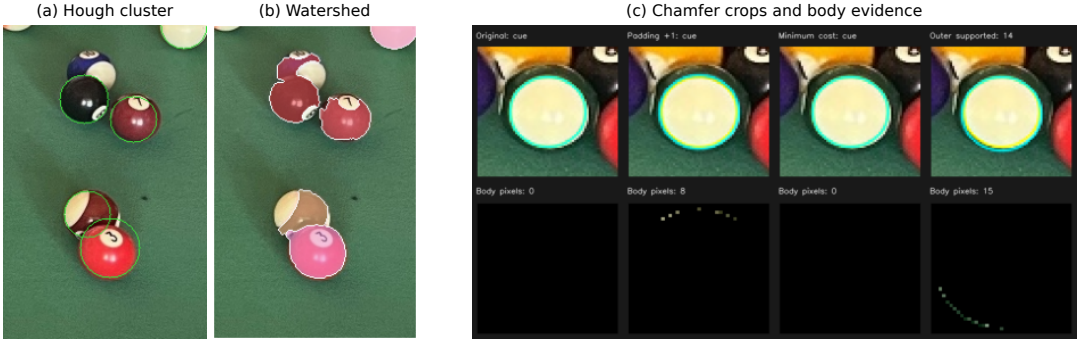


Figure 7: Watershed and chamfer examples. A plausible region boundary does not necessarily improve the final detection or identity.

Ground truth and generalisation

Annotations supply centres and identities, not visible masks or true image radii. They cannot establish whether a segmentation boundary is correct. Occlusion and shadows make some centres difficult to judge; no specific annotation has been shown incorrect. The homography describes the cloth plane, whereas balls extend above it. The 15-unit tolerance is about 35% of the 43 mm ball diameter, so checking other tolerances would help assess the results.

The fixed camera, narrow radius range and limited lighting variation favour this system. AllBalls covers only rows 143–197, making much of the radius model an extrapolation. Cue reservation can fail if a striped patch is larger or the real cue was not initially labelled white. Repeated positions and a shared viewpoint also limit the diversity of the six images. More varied training views and new test scenes are needed. The observed 80/86 is promising, but does not demonstrate the same accuracy under different conditions.

Appendix A: final results for all six tables

These are the final rectified views from the selected configuration, using the same parameters for every image. Labels show predictions, including Unknown; they have not been replaced with the annotated identities. Positions use the original Hough centres. Individual full-resolution images accompany the report source; the overview below retains all six images for comparison.

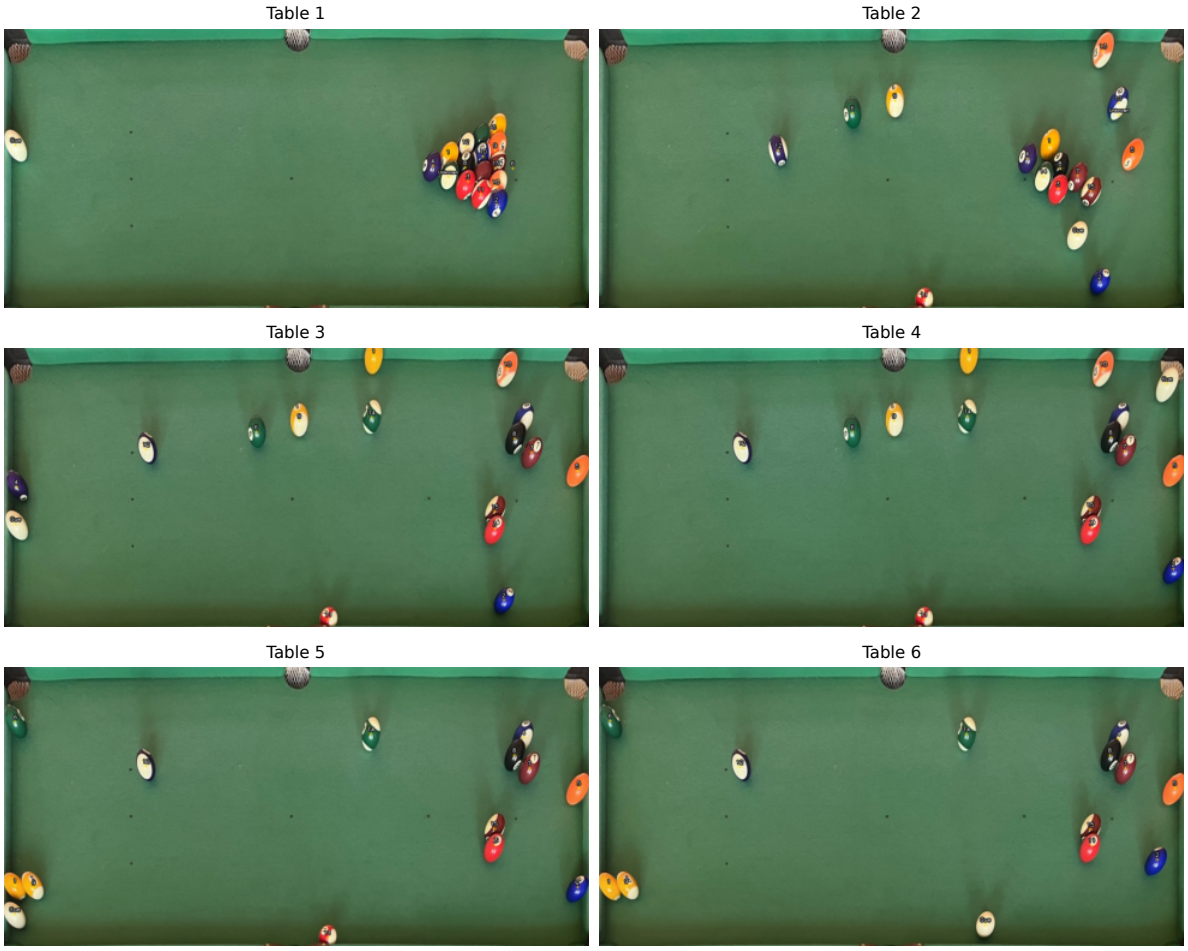


Figure 8: Final predicted identities and positions for Table1–Table6. Read together with the per-image metrics and failure examples in the main text.

All six real cue balls are correctly identified. Table1 contains three identity errors among its 14 matched detections; Table5 contains one, and Table6 two. Tables2–4 correctly identify every matched detection, although Tables3 and 4 each miss one ball. The extra Table2 detection remains Unknown and still counts as a false positive.

Appendix B: radius measurements and resources

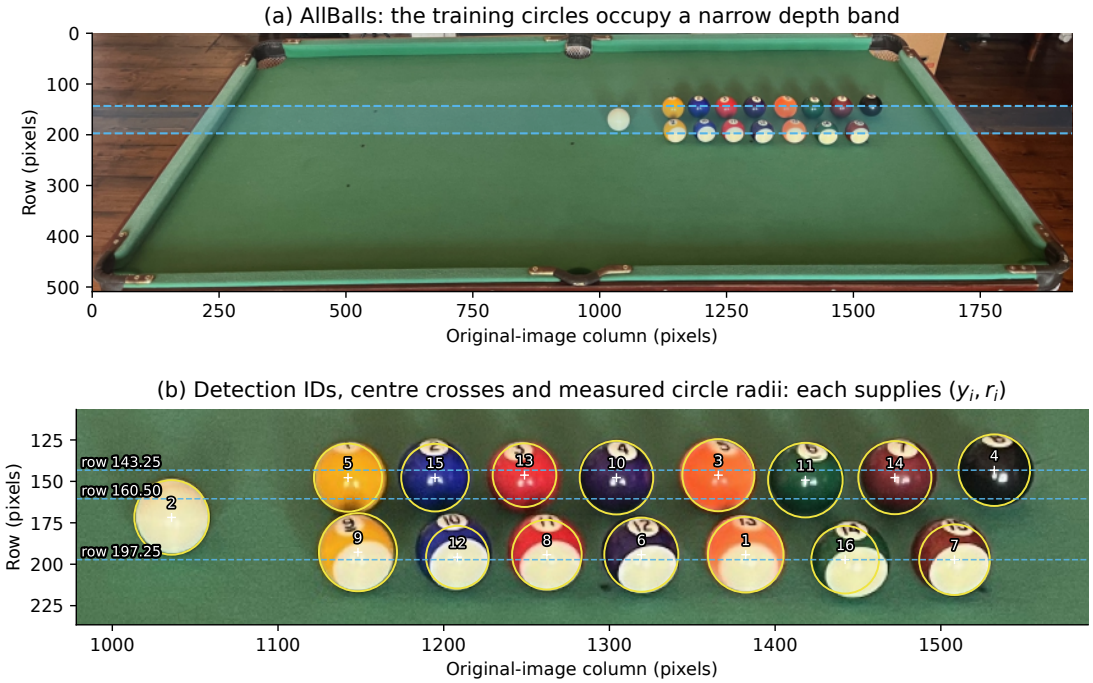


Figure 9: The AllBalls centre-row band and enlarged measured circles. Crosses mark centres; numbers identify detections rather than ball identities. Circles supply the row/radius pairs used in the model.

The slope is the median of 71 pairwise slopes with centre rows at least ten pixels apart. The intercept is the median of the 16 values $r_i - by_i$. Medians reduce the influence of unusually large or small detections, but cannot remove a shared detection bias or create evidence at unobserved depths.

Resources and generative AI acknowledgement

Images, dimensions, corners and annotations are supplied module materials. OpenAI ChatGPT/Codex assisted with design, C++ experiment/evaluation code, debugging, comparisons, figure scripts, diagrams and this report's prose. AI-generated contributions include histogram, identity, padding, cue-size and alternative-method experiments, numerical summaries and draft illustrations.

References

[1] M. Swain and D. Ballard. Color indexing. *IJCV* 7, 11–32 (1991). doi:10.1007/BF00130487.

[2] H. Kuhn. The Hungarian method for the assignment problem. *Naval Research Logistics Quarterly* 2, 83–97 (1955).

[3] P. Sen. Estimates of the regression coefficient based on Kendall's tau. *JASA* 63, 1379–1389 (1968).

[4] L. Vincent and P. Soille. Watersheds in digital spaces. *IEEE TPAMI* 13, 583–598 (1991).

[5] G. Borgefors. Hierarchical chamfer matching: a parametric edge matching algorithm. *IEEE TPAMI* 10, 849–865 (1988).